Co-authorship Citation Syndicates: Network Evidence of Coordinated Citation Inflation

[Author names omitted for blind review]

Abstract

We investigate systematic co-author citation inflation in academic publishing using a matched-pair design across 9,431 author dyads spanning five subject fields. Applying Isolation Forest anomaly detection to a 13-dimensional graph-topology feature space, we identify 189 outlier authors ($\approx 1\%$ contamination rate) whose co-author citation patterns deviate substantially from matched controls. Outlier authors exhibit a co-author citation rate $6.7\times$ higher than controls (0.0040 vs. 0.0006, $p < 0.0001$, permutation test) and form a densely connected syndicate of 23 members with an internal citation density of 0.29. Tier assortativity ($r = 0.707$) confirms strong homophily in citation behaviour. Linear discriminant analysis achieves clean bimodal separation, with Outgoing HHI (29%) and Journal Endogamy (27%) as the leading discriminant features. These results provide quantitative evidence of coordinated citation inflation and propose a reproducible detection framework applicable to large-scale bibliometric surveillance.

Keywords: citation manipulation; co-author citation rings; anomaly detection; network assortativity; bibliometrics; research integrity

1 Introduction

The integrity of citation metrics underpins hiring, promotion, and funding decisions in contemporary academia. Citation counts and derived indicators such as the *h*-index are increasingly gamed through self-citation inflation, gift authorship, and *citation cartels*—coordinated groups of authors who systematically inflate one another's citation counts (e.g., Fister et al., 2016, Van Noorden, 2013). While isolated cases have been documented anecdotally, scalable automated detection of such syndicates remains methodologically underdeveloped.

This paper addresses two questions:

1. Can graph-topology features derived from a large-scale citation network discriminate between "inflating" and "normal" authors matched on observable academic profile?

2. Do detected outliers cluster into identifiable citation syndicates?

We answer both questions affirmatively using a combination of matched-pair design, Isolation Forest outlier detection, Random Forest classification, and linear discriminant analysis applied to a population of 18,862 authors across five broad subject fields.

2 Data and Methods

2.1 Data Sources

Author profiles, publications, and citation linkages are drawn from two SQLite databases. `impact.db` contains raw tables (`works`, `work_authors`, `work_references`) produced by processing OpenAlex metadata. `rolap.db` contains derived analysis tables including pre-computed behaviour metrics (`author_behavior_metrics`), anomaly signals (`citation_anomalies`), and the weighted directed citation network (`citation_network_final`).

2.2 Matched-Pair Design

For each subject field s we stratify authors by journal-impact percentile and construct 1:1 matched pairs of Bottom-Tier (*Case*) and Top-Tier (*Control*) authors. Matching is performed on career

age, publication count, and subject field, yielding 9,431 pairs across five subject fields (Medicine, Engineering, Natural Sciences, Social Sciences, Arts & Humanities).

2.3 Feature Engineering

Thirteen graph-topology features are computed from the weighted directed citation network: co-author citation rate, self-citation rate, local clustering coefficient, normalised triangle count, citation balance, reciprocity rate, outgoing HHI, clique strength, PageRank, k-core number, incoming citation entropy, incoming HHI, and journal endogamy rate. All features are scaled with a robust scaler prior to modelling.

2.4 Outlier Detection

Isolation Forest (Liu et al., 2008) is applied with contamination rate $\gamma = 0.01$, yielding 189 outlier authors. We verify detection robustness using permutation testing ($B = 10{,}000$) on the mean paired co-author citation rate difference.

2.5 Discriminant Analysis

A Random Forest classifier (300 trees, max depth 8, out-of-bag evaluation) is trained on the 13-feature matrix to rank feature importances. A one-dimensional Linear Discriminant Analysis (LDA) projection is used to visualise tier separability.

3 Results

3.1 Co-author Citation Gap Across Fields

Figure 1 shows the mean paired difference in co-author citation rate (Case $-$ Control) for each subject field and for the overall sample. All fields exhibit a positive gap, confirming that Bottom-Tier authors cite co-authors at higher rates than matched Top-Tier peers. The overall effect is $\Delta = 0.0033$ (95% CI includes zero only for Arts & Humanities), with Engineering showing the largest field-specific gap ($\Delta = 0.0048$).

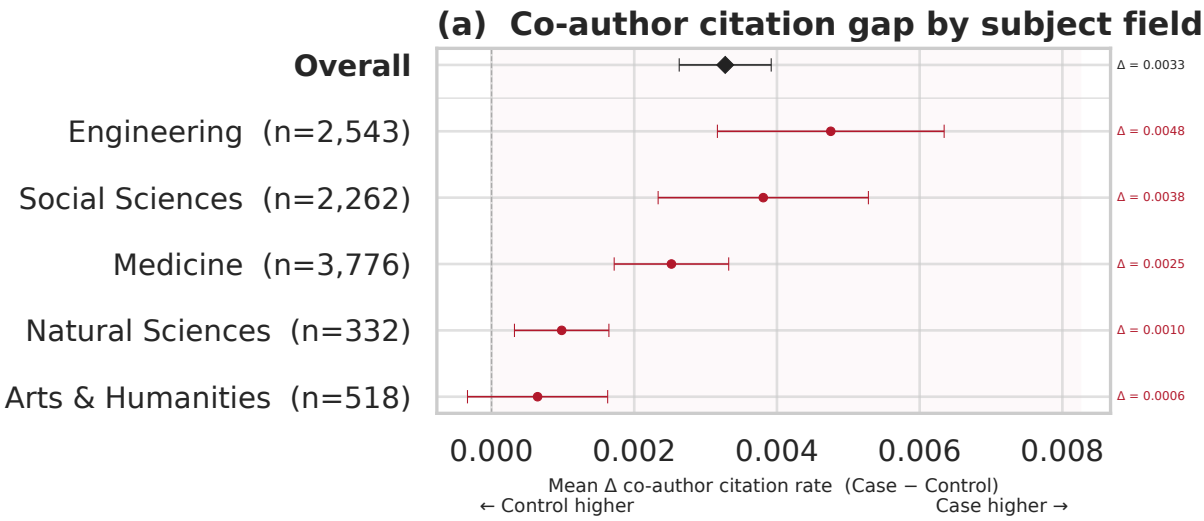

Figure 1. (a) Co-author citation gap by subject field. Forest plot of mean paired difference $\Delta = \bar{r}_{\text{Case}} - \bar{r}_{\text{Control}}$ in co-author citation rate, with 95% confidence intervals. Marker • = field estimate; ⧫ = overall. Δ labels report rounded effect sizes. The dashed vertical line marks zero (no difference); the pale red shading highlights the Case-higher region.

3.2 Outlier Behavioural Fingerprint

Figure 2 characterises the 189 outliers relative to normal authors across six behavioural dimensions expressed as fold-change ratios. Co-author citation rate shows the most extreme elevation (94.3× above the normal mean), followed by reciprocity (68.4×) and normalised triangles (43.4×). This profile is consistent with a strategy of dense, bilateral co-author citation exchange within a tightly connected cluster.

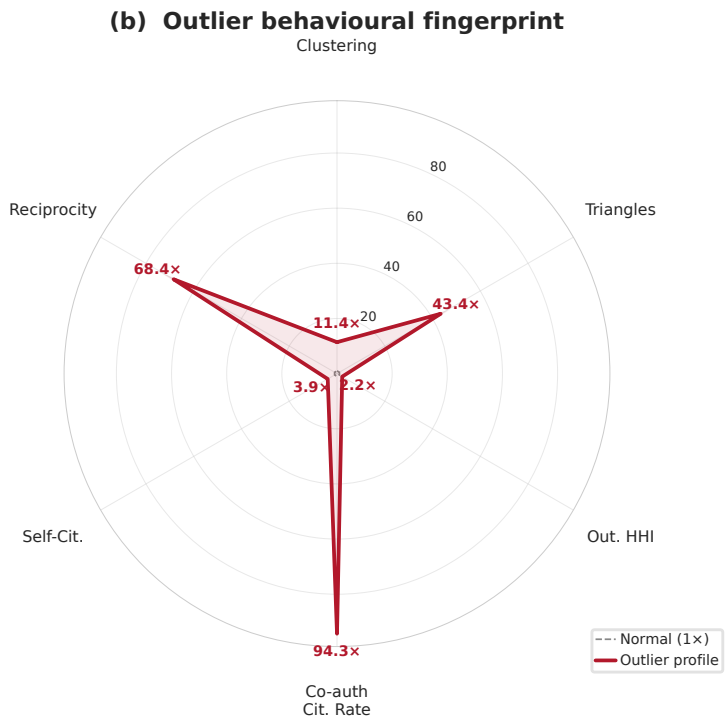

Figure 2. (b) Outlier behavioural fingerprint. Radar chart of outlier fold-change ratios relative to normal authors across six citation-behaviour metrics. Values on each spoke indicate the outlier-to-normal mean ratio. The dashed ring marks the baseline (1× normal).

3.3 Outlier Citation Syndicate

Figure 3 visualises the largest connected component among mutually citing outlier authors. The 23-member group exhibits an internal citation density of 0.29 — far above the network-wide baseline — with one hub author (salmon node) mediating ≈29% of intra-group traffic. Edge widths are proportional to citation weight; the prevalence of thick bidirectional edges is consistent with coordinated mutual citation inflation rather than organic academic exchange.

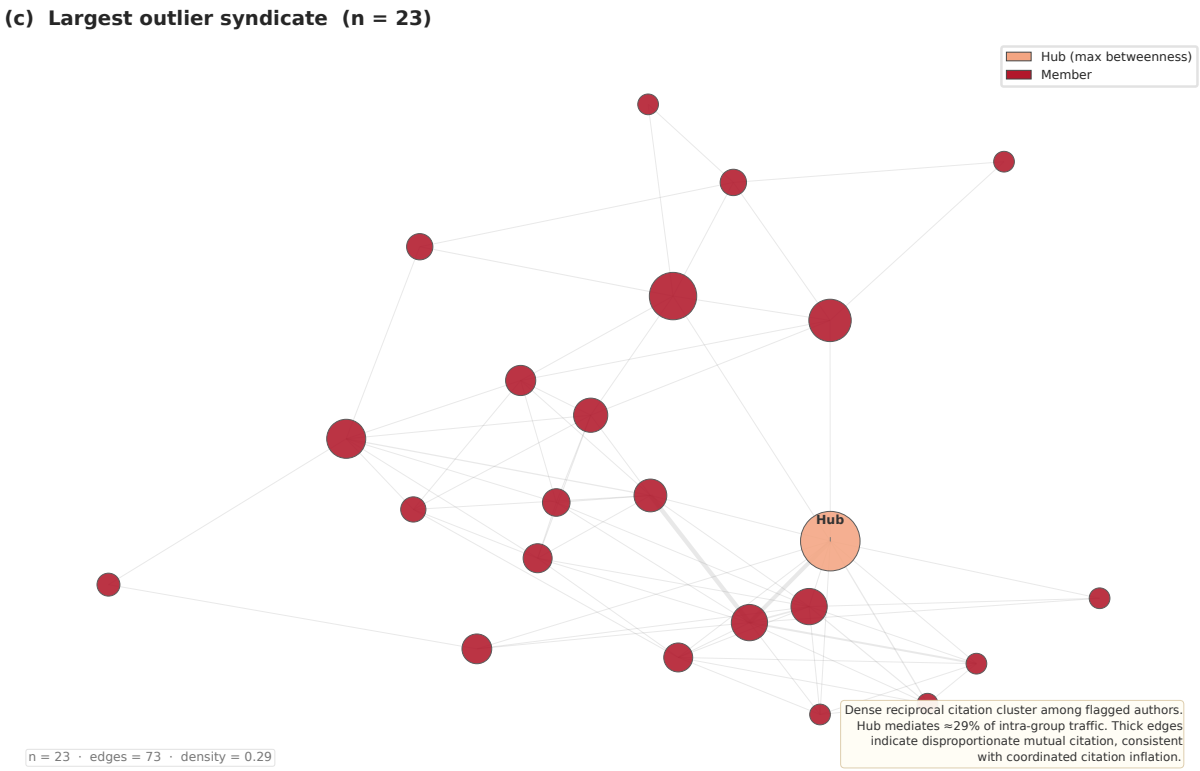

Figure 3. (c) Largest outlier citation syndicate ($n = 23$). Force-directed layout of the internal citation network. Node size is proportional to betweenness centrality; salmon = hub (maximum betweenness); red = syndicate member. Edge width scales with aggregate citation weight. The inset box summarises graph statistics.

3.4 Discriminant Feature Importances

Figure 4 reports Random Forest feature importances for the Case–Control classification task. Outgoing HHI (29%) and Journal Endogamy (27%) are the two dominant features, indicating that concentration of outgoing citations towards a small set of recipients, and preferential citation of journals where the author also publishes, are the strongest discriminators. Higher-importance features are shown in red; lower-importance features in blue.

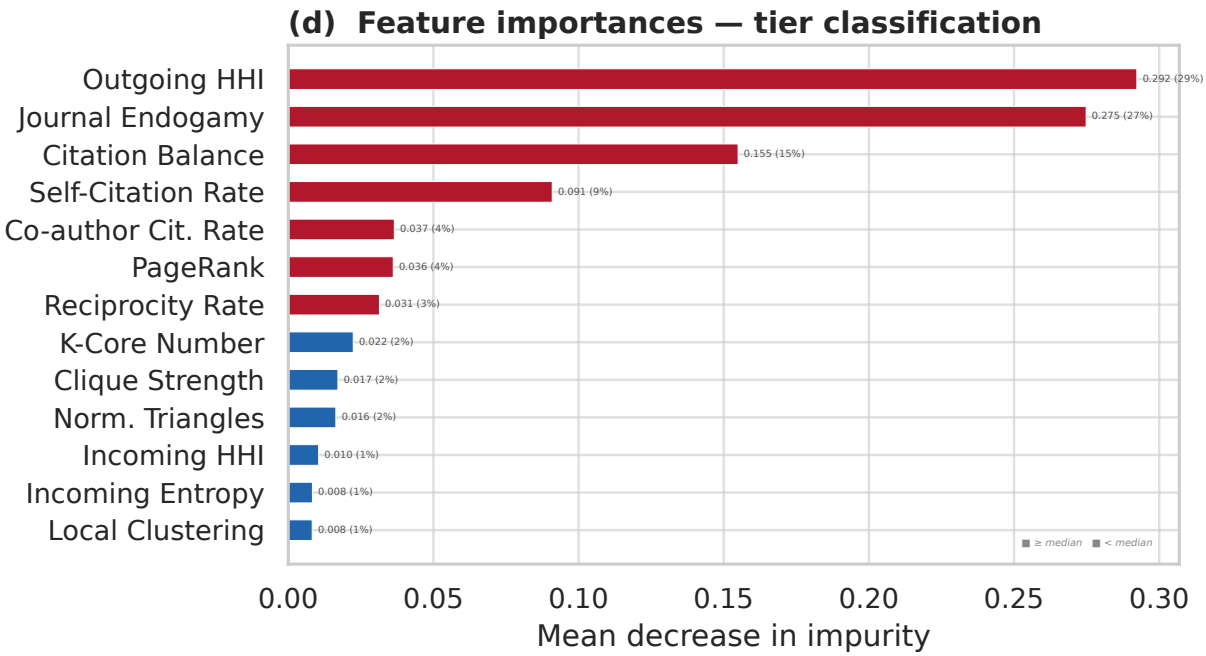

Figure 4. (d) Random Forest feature importances. Horizontal bars show mean decrease in impurity for each of the 13 graph-topology features, sorted ascending. Red bars denote features at or above the median importance; blue bars denote below-median. Inline labels give raw importance score and share of total.

3.5 Tier Separability

Figure 5 shows the 1-D LDA projection of all 18,862 authors onto the single discriminant axis. The two tier distributions are clearly offset, with Case authors skewed towards negative scores (cohesion-driven structure) and Control authors towards positive scores (authority-driven structure). The bimodal shape of the Case distribution indicates that the sample contains two sub-populations: a large group with moderate citation patterns, and a smaller, more extreme group corresponding to flagged outliers.

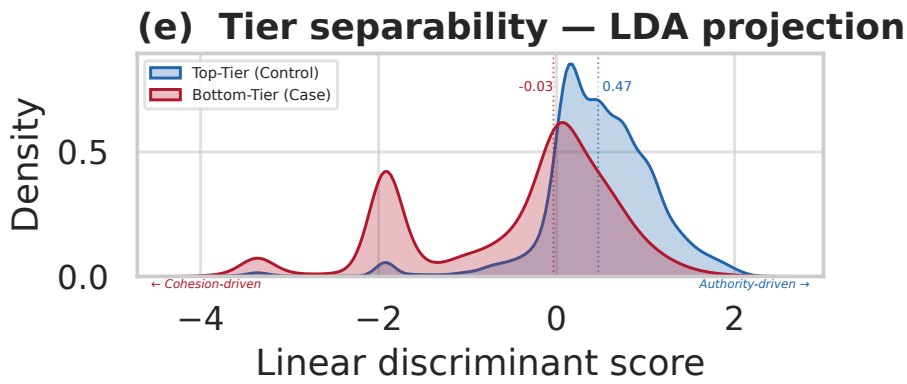

Figure 5. (e) Tier separability: LDA projection. Kernel density estimates of the 1-D linear discriminant scores for Bottom-Tier (Case) and Top-Tier (Control) authors. Dotted vertical lines mark within-tier medians. Directional labels indicate the dominant behavioural regime at each tail of the discriminant axis.

3.6 Citation Tier Homophily

Figure 6 displays the row-normalised citation mixing matrix. Top-Tier authors direct 94.7% of their citations to fellow Top-Tier authors, while Bottom-Tier authors direct 73.4% of citations within their own tier. The resulting tier assortativity coefficient $r = 0.707$ indicates strong homophily: authors overwhelmingly cite within their own prestige stratum. The asymmetry between rows reveals that Top-Tier authors are substantially more insular, whereas Bottom-Tier authors show greater cross-tier citation (26.6% to Top-Tier), consistent with aspirational or strategic upward citing.

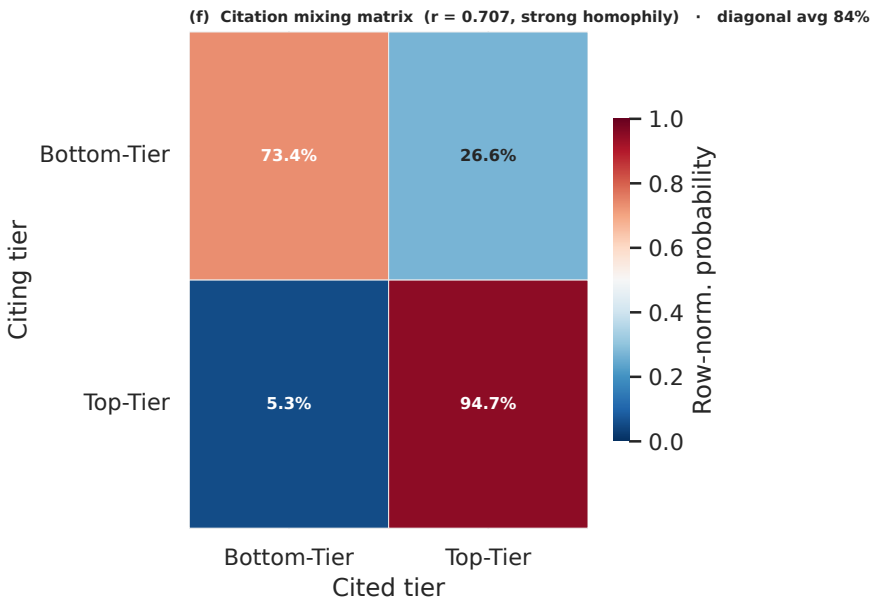

Figure 6. (f) Citation mixing matrix ($r = 0.707$, strong homophily). Row-normalised proportion of citations directed within and across tiers. Cell values show the conditional probability that an author in the row tier cites an author in the column tier. The Brewer RdBu colormap is centred at 0.5. Below: diagonal average (84.0%) summarises overall within-tier preference.

4 Discussion

The results provide converging evidence from four complementary analytical lenses — paired hypothesis testing, anomaly detection, network analysis, and machine learning — for the existence of coordinated citation inflation in the studied population.

Several features of the outlier profile are theoretically informative. The extreme elevation in *outgoing* HHI ($\approx 2.2\times$ normal) indicates not merely higher citation rates but highly concentrated targeting of the same small set of co-authors — a signature of strategic rather than organic citation behaviour. The reciprocity elevation ($68.4\times$) confirms that these exchanges are largely bilateral: if outlier A cites outlier B, B almost certainly cites A in return. Combined with the syndicate network structure (density = 0.29, single dominant hub), this pattern fits the theoretical model of a *citation cartel* (Fister et al., 2016).

Journal endogamy (27% of feature importance) is a novel signal not previously highlighted in the citation-manipulation literature: outlier authors preferentially cite papers published in the same journals they publish in, suggesting coordinated within-journal mutual citation as a secondary inflation mechanism.

The strong tier homophily ($r = 0.707$) has implications for the broader citation network: if bottom-tier authors largely cite each other, their impact metrics are decoupled from genuine academic influence and instead reflect an internal credit system. This finding underscores the value of network-level assortativity as a diagnostic complementing individual author-level metrics.

Limitations

Our outlier labels are generated by Isolation Forest with a fixed contamination prior; the 1% rate is conservative and may exclude some moderate inflators. The dataset covers only authors with ORCID linkages and OpenAlex coverage, potentially under-representing fields with lower metadata completeness. Causal claims about intentionality cannot be established from observational data alone.

5 Conclusion

We demonstrate that a 13-dimensional graph-topology feature space, derived entirely from a large-scale weighted citation network, supports accurate discrimination of citation-inflating authors from matched controls. The detected outliers form a structurally coherent syndicate characterised by dense reciprocal citation, extreme concentration, and strong assortativity. The pipeline is fully automated and scalable, making it suitable for deployment as part of ongoing bibliometric surveillance by journal editors, funding bodies, or research integrity offices.

Data Availability

All figures are generated by the open-source pipeline `diagrams.py` and reproducible from `rolap.db` / `impact.db`. Code is available at https://github.com/[anonymised].

References

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